Trap-limited and percolation conduction mechanisms in amorphous oxide semiconductor thin film transistors

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The electron conduction mechanism in the above-threshold regime in amorphous oxide semiconductor thin film transistors is shown to be controlled by percolation and trap-limited conduction. The band tail state slope controls the field effect mobility, while the average spatial coherence length and potential fluctuation control percolation conduction. In these limits, the field effect mobility is found to follow a power law, from which a universal mobility versus carrier concentration dependence is extracted.

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REFERENCES

1. H. Q. Chiang, D. Hong, C. M. Hung, R. E. Presley, J. F. Wager, C. Park, D. A. Keszler, and G. S. Herman, J. Vac. Sci. Technol. B **24**, 2702 (2006).
2. A. Suresh, P. Wellenius, A. Dhawan, and J. Muth, Appl. Phys. Lett. **90**, 123512 (2007). [ISI]
3. K. Ghaffarzadeh, A. Nathan, J. Robertson, S. Kim, S. Jeon, C. Kim, U. Chung, and J. Lee, Appl. Phys. Lett. **97**, 113504 (2010).
4. K. Nomura, T. Kamiya, H. Ohta, K. Ueda, M. Hirano, and H. Hosono, Appl. Phys. Lett. **85**, 1993 (2004).
5. J. Robertson, Phys. Status Solidi B **245**, 1026 (2008).
6. S. Jeon, S. I. Kim, S. Park, I. Song, J. Park, S. Kim, and C. Kim, IEEE Electron Device Lett. **31**, 1128 (2010).
7. C. Lee, B. Cobb, and A. Dodabalapur, Appl. Phys. Lett. **97**, 203505 (2010).
8. T. Kamiya, K. Nomura, and H. Hosono, Appl. Phys. Lett. **96**, 122103 (2010).
9. T. Kamiya, K. Nomura, and H. Hosono, J. Disp. Technol. **5**, 462 (2009).
10. P. Servati, D. Striakhilev, and A. Nathan, IEEE Trans. Electron Devices **50**, 2227 (2003).
11. S. Kirkpatrick, Rev. Mod. Phys. **45**, 574 (1973).
12. S. M. Sze, *Physics of Semiconductor Devices*, 2nd ed. (Wiley, Hoboken, NJ, 1981).
13. W. B. Joyce and R. W. Dixon, Appl. Phys. Lett. **31**, 354 (1977).
14. K. Jeon, C. Kim, I. Song, J. Park, S. Kim, S. Kim, Y. Park, J. Park, S. Lee, D. M. Kim, and D. H. Kim, Appl. Phys. Lett. **93**, 182102 (2008).
15. A. Cerdeira, M. Estrada, R. Garcıa, A. Ortiz-Conde, and F. Garcıa Sánchez, Solid-State Electron. **45**, 1077 (2001).

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Oxide semiconductors exhibit high electron mobility even when fabricated at room temperature, making them the leading candidate for next generation thin film transistor (TFT) technology.[1–3] This class of materials have unique carrier transport mechanisms despite sharing similar attributes of other TFT technologies.[4] Contrary to hydrogenated amorphous silicon (a-Si:H), their conduction band edge (E_m) is composed of spherical overlapping orbitals, leading to a reduced density of band tail states in the amorphous phase, thus eliminating Fermi level (E_F) pinning in TFTs.[4,5] Furthermore, recent experiments suggest that carriers percolate between potential barriers in nonlocalized states around E_m.[5] These properties elude to a carrier transport picture governed by both trap-limited and percolation mechanisms in oxide semiconductor TFTs.

In this letter, the relative dominance of trap-limited conduction (TLC) and percolation is investigated in bilayer a-InZnO/GaInZnO TFTs. A bilayer structure is employed to combine high band mobility of In–Zn–O with the positive threshold voltage (V_T) of In–Ga–Zn–O.[6] Based on the above-threshold transfer characteristics, we deduce that in the limit of low gate voltage, TLC prevails.[7] In contrast, at high gate voltages, percolation becomes dominant.[7,8] In both cases, the field effect mobility (μ_{FE}) follows a power law. In particular, percolation conduction is determined by the potential barrier height and the average barrier width (W_B)-to-distance (D_B) coherence ratio. The value of the average potential barrier height (ϕ_{B0}) and variance (σ_B) are consistent with earlier reports.[7–9] At high electric fields, we observe slight degradation in μ_{FE} which we believe is most likely due to surface scattering.

Figure 1 illustrates a conceptual band diagram of a typical oxide semiconductor TFT, depicting an exponential distribution of localized tail states as well as potential barriers above E_m.[8] When the Fermi level (E_F) resides within the localized tail states ($E_F \ll E_m$), carrier transport is controlled by multiple trap and release events, i.e., by TLC. Here, μ_{FE} is proportional to $n_{free}/(n_{free}+n_{trap})$, where n_{free} and n_{trap} are the free and trapped carrier densities, respectively. The usual band mobility prefactor that appears is now modulated by a percolation term as shown below. When $T_t > T$, the TLC-dominated μ_{FE} for $E_F < E_m$ can be analytically approximated as a power law[10]

$$\mu_{FE} = \mu_0 \exp\left[-\frac{q\phi_{B0}}{kT} + \frac{(q\sigma_B)^2}{2(kT)^2}\right] A^*(V_{GS} - V_T)^{2(T_t/T-1)}. \quad (1)$$

Here, A^* is related to the density of tail states (N_{tc}) and T_t and V_T denote the characteristic temperature of band tails states and threshold voltage of the TFT, respectively.[10] Note that the band mobility (μ_0) is modulated by a percolation term, $\exp[-q\phi_{B0}/kT + (q\sigma_B)^2/2(kT)^2]$, derived from the Thomas–Fermi approximation.[9]

In contrast, when E_F lies between E_m and E_M, the tail states are filled therefore μ_{FE} is solely controlled by percolation. Here, carriers released into the nonlocalized states move by finding the path of least resistance among the potential barriers (see Fig. 1).[11] In this limit, gate voltage-induced band bending ($\Delta\varphi$) changes the barrier height at the interface since this is given by $E_M - E_F$, i.e., $q(\phi_{B0} - \Delta\varphi)$. Therefore,

$$\mu_{FE} = \mu_0 \exp\left[-\frac{q\phi_{B0}}{kT} + \frac{(q\sigma_B)^2}{2(kT)^2}\right] \exp\left(\frac{q\Delta\varphi}{kT}\right). \quad (2)$$

When E_F exceeds E_m, carrier density has no temperature dependence,[8] which implies that the carrier density is degen-

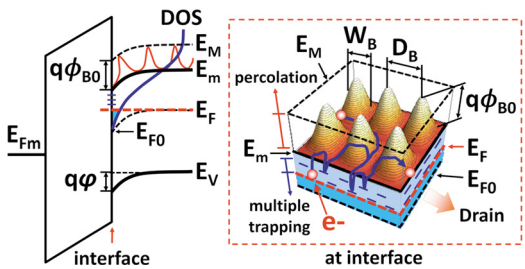


FIG. 1. (Color online) Schematic band diagram of an oxide semiconductor TFT showing the presence of band tail states and potential fluctuations above E_m, giving rise to TLC and percolation.

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erate. However, it is difficult to represent free carrier density analytically when E_F is in the vicinity of E_m.[12] In order to represent free carrier density analytically when E_F is at the vicinity of E_m, the relation, $\exp[(E_F-E_m)/kT]=(n_{free}/N_C)+2^{-3/2}(n_{free}/N_C)^2$ based on a numerical approach, is employed,[13] where N_C is the effective density of states. Solution of this relation yields approximately $n_{free} \approx 2^{3/4}N_C \exp[(E_F-E_m)/2kT]$ for $1<\sqrt{2}\exp[(E_F-E_m)/kT]$. Based on this relation, the surface electric field (E_S), which is function of $\Delta\varphi=(E_F-E_m)/q$, can be analytically represented by $\Delta V_{GS}=V_{GS}-V_p$ via Poisson's equation, assuming $n_{free} \gg n_{trap}$ when $E_M>E_F>E_m$ (Ref. 12)

$$E_S \approx \sqrt{\frac{2q}{\varepsilon_S}\int_0^{\Delta\varphi}\left[2^{3/4}N_C \exp\left(\frac{q\Delta\varphi'}{2kT}\right)\right]d\Delta\varphi'} \approx \frac{C_i(V_{GS}-V_p)}{\varepsilon_S}. \quad (3)$$

Solution of Eq. (3) yields an analytical relationship between $\Delta\varphi$ and ΔV_{GS}

$$\Delta\varphi \approx \frac{2kT}{q}\ln\left[\frac{C_i^2(V_{GS}-V_p)^2}{2^{11/4}\varepsilon_S N_C kT}\right]. \quad (4)$$

Here, V_p is a percolation threshold voltage when $E_F=E_m$ and ε_S relative permittivity of the channel layer.

In arriving at Eq. (4), the conduction band profile was assumed to be spatially flat. However, as mentioned earlier, the conduction band contains potential barriers which can be spatially characterized using an average barrier width (W_B) and distance (D_B) (see Fig. 1). This band profile of oxide semiconductors renders $\Delta\varphi$ dependent on W_B and D_B, because when $E_M>E_F>E_m$, charge is primarily induced within areas adjacent to these barriers. Therefore, to maintain charge neutrality, a larger $\Delta\varphi$ is required for the same ΔV_{GS} when the barriers are wider. Consequently, $\Delta\varphi$ becomes modified by the spatial property of the barriers, $\Delta\varphi[D_B/(D_B-W_B)]$. Based on Eqs. (2) and (4), the gate voltage dependent μ_{FE} for $E_M>E_F>E_m$ can be approximated analytically

$$\mu_{FE}=\mu_0 \exp\left[-\frac{q\phi_{B0}}{kT}+\frac{(q\sigma_B)^2}{2(kT)^2}\right]B^*(V_{GS}-V_P)^{4\left(\frac{D_B-W_B}{D_B}\right)}, \quad (5)$$

where B^* is $[C_i^2/(2^{11/4}\varepsilon_S N_C kT)]^{2[(D_B-W_B)/D_B]}$. Revisiting Eqs. (1) and (5), the gate voltage dependent mobility in the limits ($E_F \ll E_m$) and ($E_M>E_F>E_m$), where TLC and percolation dominate, respectively, can be generalized using a power law

$$\mu_{FE}=K(V_{GS}-V_{T,P})^{\gamma}, \quad (6)$$

with the appropriate values of K and γ.[14] Equation (6) together with the measured values shown in Fig. 2 yield values for K_1 and γ_1 of 7.6 and 0.7, respectively, in the limit where TLC is dominant and K_2 and γ_2 of 38.9 and 0.1, respectively, in the limit where percolation is dominant. A schematic cross-section of the TFTs examined here is depicted as an inset in Fig. 2. Details of the device fabrication have been reported elsewhere.[6] The TFTs have the following geometrical and physical parameters: channel aspect ratio 25 μm/100 μm, subthreshold slope 0.36 V/dec, threshold voltage (V_T) 2.5 V, gate capacitance (C_i) 3.54 $\times 10^{-8}$ F/cm^2, and contact resistance (R_{SD}) 4 kΩ. As

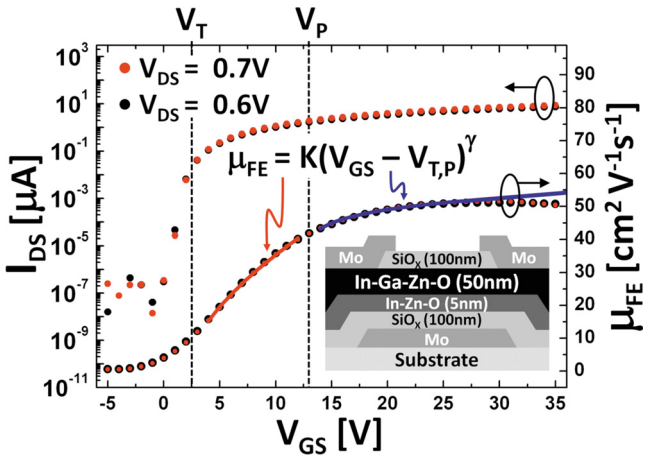


FIG. 2. (Color online) Drain-source current (I_{DS}) and field effect mobility (μ_{FE}) as a function of V_{GS} with two different values of drain voltage (V_{DS} =0.6 V, 0.7 V); red and blue lines depict power law fits to μ_{FE} (V_{GS}) at the limits of when TLC and percolation dominates, respectively. Inset shows a schematic cross-section of the TFT with In–Zn–O/Ga–In–Zn–O bilayer channel (Ref. 6).

shown in Fig. 2, the thickness of In–Zn–O layer (5 nm) is ten times thinner than that of Ga–In–Zn–O layer (50 nm) and much thinner than Debye length (>120 nm) when film doping concentration is less than 10^{15} cm^{-3}. This suggests that the entire channel layer is fully accumulated. Hence, we assume that conduction is largely in the Ga–In–Zn–O layer rather than the In–Zn–O layer.

While we have the mobility behavior in the limits of TLC and percolation, exactly where the transition lies between the two limits is fuzzy and depends on the TFT process and the material system used. For the devices considered here, the transition point can be taken as percolation threshold (V_p), which is the value of V_{GS} at which $E_F=E_m$. In order to estimate this transition point accurately, we compute n_{free} and n_{trap} using Fermi–Dirac statistics (see Fig. 3) (Refs. 12)

$$n_{free}(E_F)=\left[\frac{(2m_n^*)^{3/2}}{2\pi^2\hbar^3}\right]\int_{E_m}^{\infty}\sqrt{E-E_m}f(E)dE, \quad (7)$$

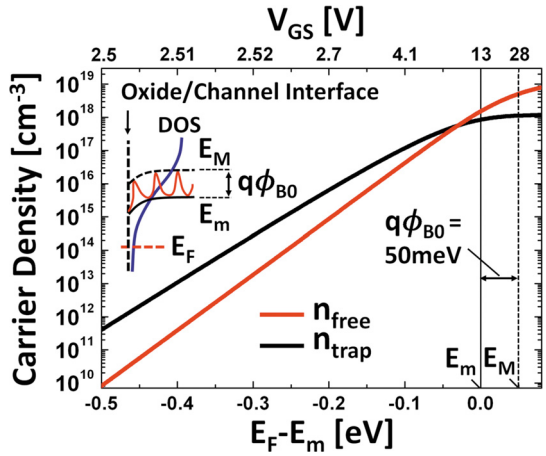


FIG. 3. (Color online) Carrier densities (n_{free} and n_{trap}) as a function of Fermi level energy (E_F) from the mobility edge (E_m); inset shows schematic band diagram at the SiO_x/channel (In–Zn–O/In–Ga–Zn–O) interface.

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TABLE I. Material parameters (kT_t, N_{tc}, and $n_{free}/(n_{free}+n_{trap})$ extracted when E_F-E_m is -0.2 eV) retrieved for the a-InZnO/GaInZnO TFTs, examined here, in comparison with the conventional a-Si:H TFTs from the literatures.

Parameter	a-InZnO/GaInZnO	a-Si:H (16,17)
μ_0 ($cm^2\ V^{-1}\ s^{-1}$)	~70 (15)	~10
kT_t (eV)	0.035	0.031
N_{tc} ($cm^{-3}\ eV^{-1}$)	$\sim 4\times10^{19}$	$\sim 5\times10^{22}$
$n_{free}/(n_{free}+n_{trap})$	0.2	0.004

$$n_{trap}(E_F) = \int_{E_{F0}}^{E_m} N_t(E) f(E) dE, \tag{8}$$

where $N_t(E)=N_{tc}\exp[(E-E_m)/kT_t]$, $f(E)=[1+\exp\{(E-E_F)/kT\}]^{-1}$, and N_{tc} is the density of trap states at $E=E_m$.[10] Through Poisson's equation, the transition voltage (V_p) can be estimated using Eqs. (7) and (8) (Ref. 12)

$$V_p \approx \frac{\sqrt{2q\varepsilon_S \int_{E_{F0}}^{E_m}[n_{free}(E_F)+n_{trap}(E_F)]dE_F}}{C_i}. \tag{9}$$

By using Eqs. (1), (5), and (6), N_{tc} and ϕ_{B0} can be computed using the values of K_1 and K_2 in both conduction limits. Assuming that the effective mass of bilayer structure computed using a harmonic average between $0.15m_0$ for In–Zn–O and $0.35m_0$ for Ga–In–Zn–O is $0.21m_0$ and band mobility is 70 $cm^2/V^{-1}\ s^{-1}$,[15] the extracted value of N_{tc} is about 4×10^{19} $cm^{-3}\ eV^{-1}$ when ϕ_{B0} is 50 meV with σ_B of 45 meV assuming a Gaussian distribution of the potential barriers. The value of N_{tc} is about three orders of magnitude smaller than that of a-Si:H (see Table I).[16,17] Based on the extracted values of N_{tc} and ϕ_{B0}, the transition point computed by using Eqs. (7)–(9) yields $V_p=13$ V.

When V_{GS} is decreased below V_p of 13 V, n_{trap} becomes an order of magnitude larger than n_{free}. In this region, the dominance of TLC grows, yielding T_t of 403 K from the extracted value of γ_1 based on Eqs. (1) and (6) (see Fig. 2). In particular, when $E_F-E_m=-0.2$ eV, corresponding to the threshold voltage V_T, the ratio $n_{free}/(n_{free}+n_{trap})$ for a-InZnO/GaInZnO TFTs is about 0.2, suggesting that TLC prevails.

In contrast, when V_{GS} is increased more than V_p, n_{free} becomes larger than n_{trap}, and $n_{free}/(n_{free}+n_{trap})$ approaches a value greater than 0.8 when E_F reaches E_m. Moreover, when E_F exceeds E_m, all traps are filled out so that $n_{free}/(n_{free}+n_{trap})$ becomes unity, suggesting that percolation is prevalent. Consequently, γ_2 should be interpreted using Eqs. (5) and (6), yielding a $(D_B-W_B)/D_B$ of ~0.025. This is indicative of a short spatial coherence length, implying that barriers constrict carrier movement.[8]

Furthermore, we note that μ_{FE} (V_{GS}) deviates from a power law at large V_{GS} values as shown in Figs. 2 and 4. Note that we account for the effect of contact resistance, which is 4 kΩ for the device considered.[18] It is suggested that surface roughness scattering and phonon scattering degrade μ_{FE}, yielding an effective electric field dependence, $E_{eff}^{-0.04}$ (see Fig. 4).[19]

In conclusion, we have presented a technique that provides analytical insight into carrier transport mechanisms in metal oxide semiconductor TFTs, and in particular, on the relative merits of trap-limited and percolation conduction. Although the devices considered have an a-InZnO/GaInZnO bilayer, in which device parameters have been estimated using a harmonic average and assuming that conduction mechanism is largely in the Ga–In–Zn–O layer rather than In–Zn–O layer, we argue that the prevalent mechanism is ultimately dependent on the density of tail states (N_{tc}) at the conduction band edge. A high value of N_{tc} leads to Fermi level pinning, and TLC dominates carrier transport. Percolation comes into the picture for as long as the Fermi level is able to cross the conduction band edge. We believe that N_{tc} should be below $\sim10^{20}$ $cm^{-3}\ eV^{-1}$ for percolation-dominance at a reasonable range of gate voltages of $V_{GS}<20$.

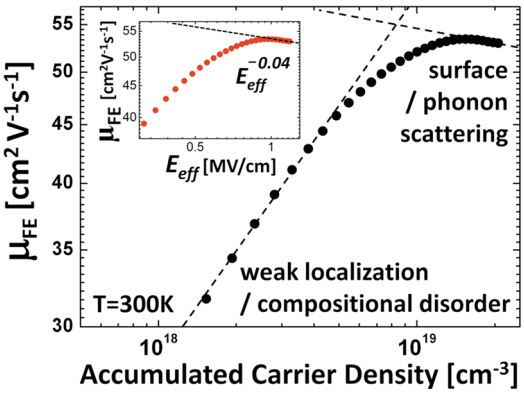


FIG. 4. (Color online) The above-threshold field effect mobility as a function of gate-induced carriers density computed from C-V measurements. Inset shows degradation of field effect mobility at 300 K on effective electric field, suggesting the mobility is limited by surface/phonon scattering (Ref. 19).

[1]H. Q. Chiang, D. Hong, C. M. Hung, R. E. Presley, J. F. Wager, C. Park, D. A. Keszler, and G. S. Herman, J. Vac. Sci. Technol. B **24**, 2702 (2006).
[2]A. Suresh, P. Wellenius, A. Dhawan, and J. Muth, Appl. Phys. Lett. **90**, 123512 (2007).
[3]K. Ghaffarzadeh, A. Nathan, J. Robertson, S. Kim, S. Jeon, C. Kim, U. Chung, and J. Lee, Appl. Phys. Lett. **97**, 113504 (2010).
[4]K. Nomura, T. Kamiya, H. Ohta, K. Ueda, M. Hirano, and H. Hosono, Appl. Phys. Lett. **85**, 1993 (2004).
[5]J. Robertson, Phys. Status Solidi B **245**, 1026 (2008).
[6]S. Jeon, S. I. Kim, S. Park, I. Song, J. Park, S. Kim, and C. Kim, IEEE Electron Device Lett. **31**, 1128 (2010).
[7]C. Lee, B. Cobb, and A. Dodabalapur, Appl. Phys. Lett. **97**, 203505 (2010).
[8]T. Kamiya, K. Nomura, and H. Hosono, Appl. Phys. Lett. **96**, 122103 (2010).
[9]T. Kamiya, K. Nomura, and H. Hosono, J. Disp. Technol. **5**, 462 (2009).
[10]P. Servati, D. Striakhilev, and A. Nathan, IEEE Trans. Electron Devices **50**, 2227 (2003).
[11]S. Kirkpatrick, Rev. Mod. Phys. **45**, 574 (1973).
[12]S. M. Sze, *Physics of Semiconductor Devices*, 2nd ed. (Wiley, Hoboken, NJ, 1981).
[13]W. B. Joyce and R. W. Dixon, Appl. Phys. Lett. **31**, 354 (1977).
[14]K. Jeon, C. Kim, I. Song, J. Park, S. Kim, S. Kim, Y. Park, J. Park, S. Lee, D. M. Kim, and D. H. Kim, Appl. Phys. Lett. **93**, 182102 (2008).
[15]A. Cerdeira, M. Estrada, R. Garcıa, A. Ortiz-Conde, and F. Garcıa Sánchez, Solid-State Electron. **45**, 1077 (2001).
[16]J. G. Shaw and M. Hack, J. Appl. Phys. **64**, 4562 (1988).
[17]R. A. Street, *Hydrogenated Amorphous Silicon* (Cambridge University Press, Cambridge, 2005).
[18]A. Ahnood, K. Ghaffarzadeh, A. Nathan, P. Servati, F. Li, M. R. Esmaeili-Rad, and A. Sazonov, Appl. Phys. Lett. **93**, 163503 (2008).
[19]S. Takagi, A. Toriumi, M. Iwase, and H. Tango, IEEE Trans. Electron Devices **41**, 2357 (1994).

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